

Principles of Economics

Problem Set 0

Universidad Carlos III de Madrid | Department of Economics

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1. Linear Functions with Positive and Negative Slopes

1. Understanding Positive Slopes

a. Consider the function $y=2x+3$. Calculate the value of y when:

a)

$$x = 1$$

b)

$$x = 4$$

c)

$$x = -2$$

2. Understanding Negative Slopes

a. Consider the function $y = -1.5x + 4$. Calculate the value of y when:

a)

$$x = 2$$

b)

$$x = -1$$

c)

$$x = 3$$

2. Calculating Slopes

- Calculate the slope of the line passing through the points (2,3) and (5,7).
- Calculate the slope of the line passing through the points (0,4) and (3,2).
- Explain what the slope represents in each of the above calculations.

3. Introducing Correlation

1. Understanding Correlation

2. Correlation in Practice

- Consider the following data sets:

x	y
1	2
2	4
3	6
4	8

- Plot the data points on a graph. Draw the line that represents the relationship between these two variables.
- What does the correlation coefficient indicate about the relationship between x and y?

1) Understanding Perfect Negative Correlation

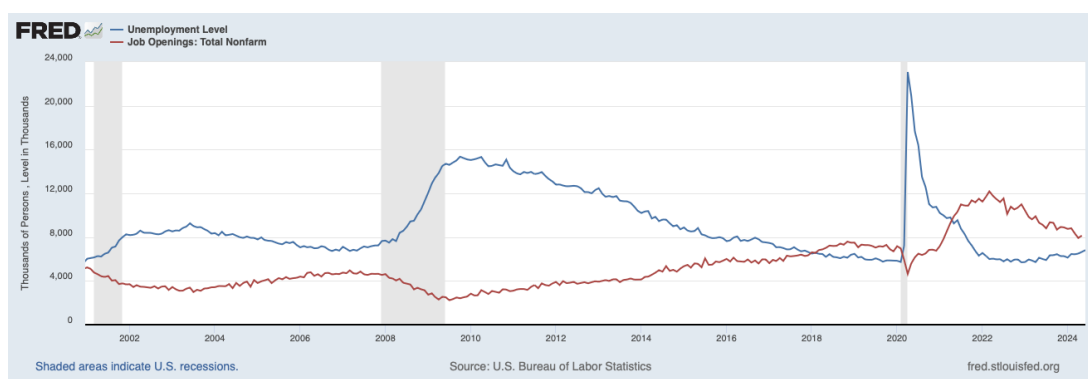
Consider the relationship between the number of hours spent on leisure activities and the number of hours spent studying for a group of students. The following table shows the data collected from 5 students:

Student	Hours of Leisure (x)	Hours of Study (y)
A	10	5
B	8	7
C	6	9
D	4	11
E	2	13

- Plot the data points on a graph with hours of leisure on the x-axis and hours of study on the y-axis. Draw the line that represents the relationship between these two variables.
- What does a correlation coefficient of -1 indicate about the relationship between these two variables?

4. Reading economic variables graphs

1. Positively correlated variables: US GDP and US Consumer spending



5. Growth Rates

- Understanding Growth Rates
- Calculating Growth Rates
- Interpreting Growth Rates

6. Calculating a Percentage of a Number

1. Calculating a Simple Percentage:

- a. A company made \$500,000 in revenue last year. This year, their revenue increased by 12%. Calculate the revenue for this year.
- b. A store is offering a 20% discount on a product that originally costs \$150. Calculate the discounted price of the product.

2. Calculating a Percentage Decrease:

- a. The population of a town decreased by 8% over the last year. If the population was 25,000 last year, what is the population this year?

3. Calculating a Percentage Increase:

- a. A student's test score increased from 75% to 85%. What is the percentage increase in the student's test score?

7. Equilibrium Price and Quantity